A DM STACK IS EXACTLY AN ALGEBRAIC STACK ADMITTING A SURJECTIVE ETALE MORPHISM FROM A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The construction of the residual gerbe at a finite type point and its local finiteness over the stack. Invoked inside the counted proof (source lines 4609-4612). Provides the DM stack at a point whose residue field will present the stack nearby.
- The existence of a surjective etale morphism from the spectrum of a field onto a one-point DM Noetherian reduced stack. Invoked inside the counted proof (source lines 4615-4617). Supplies the field and the etale map over the residual gerbe.
- The statement that an etale morphism of algebraic spaces which is universally injective is an open immersion. Invoked inside the counted proof (source lines 4440-4444 region, used at lines 4632-4636) as a cross-chapter pointer. Makes the diagonal of the comparison morphism of the presentation an open immersion.
- The slice lemma, producing locally on the presentation a closed subscheme flat and locally of finite presentation over the stack. Invoked inside the counted proof (source lines 4680-4686). Produces the closed subscheme cut out by a regular sequence near the point.
- The statement that a morphism which is flat, locally of finite presentation and unramified at a point is etale after shrinking the source. Invoked near the end of the counted proof (source lines 4707-4710). Converts the unramifiedness at the point into etaleness on a neighbourhood.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference algebra-lemma-regular-ring-CM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4365–4365. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-regular-ring-CM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebraic-definition-deligne-mumford: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4601–4601. Navigation pointer only: the counted text refers to the Stacks item algebraic-definition-deligne-mumford, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-DM-residual-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4406–4406. Navigation pointer only: the counted text refers to the Stacks item lemma-DM-residual-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-enough-finite-type-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3767–3767. Navigation pointer only: the counted text refers to the Stacks item lemma-enough-finite-type-points, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-etale-at-point: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4548–4548. Navigation pointer only: the counted text refers to the Stacks item lemma-etale-at-point, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-point-finite-type-monomorphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3803–3803. Navigation pointer only: the counted text refers to the Stacks item lemma-point-finite-type-monomorphism, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-properties-covering-imply-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1117–1117. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-covering-imply-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-separation-properties-residual-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1176–1176. Navigation pointer only: the counted text refers to the Stacks item lemma-separation-properties-residual-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-slice: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4177–4177. Navigation pointer only: the counted text refers to the Stacks item lemma-slice, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-unramified-at-point: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4689–4689. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-unramified-at-point, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-composition-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3497–3497. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-composition-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-composition-smooth: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4676–4676. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-composition-smooth, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-composition-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 880–880. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-composition-unramified, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-diagonal-unramified-morphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4660–4660. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-diagonal-unramified-morphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-locally-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4623–4623. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-locally-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines

4653–4653. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-universally-injective-open: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4656–4656. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-universally-injective-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-characterize-surjective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4141–4141. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-characterize-surjective, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-points-cartesian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2126–2126. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-points-cartesian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-topology-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2128–2128. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-topology-points, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference varieties-lemma-smooth-regular: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4682–4682. Navigation pointer only: the counted text refers to the Stacks item varieties-lemma-smooth-regular, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference varieties-lemma-smooth-separable-closed-points-dense: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4679–4679. Navigation pointer only: the counted text refers to the Stacks item varieties-lemma-smooth-separable-closed-points-dense, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Theorem 0.1. *Let $\mathcal{X}$ be an algebraic stack. The following are equivalent*

(1) *$\mathcal{X}$ is DM,*

(2) $\mathcal{X}$ *is Deligne-Mumford, and*
(3) *there exists a scheme* W *and a surjective étale morphism* $W \to \mathcal{X}$.

Proof. Recall that (3) is the definition of (2), see Algebraic Stacks, Definition algebraic-definition-deligne-mumford. The implication (3) $\Rightarrow$ (1) is Lemma lemma-properties-covering-imply-diagonal. Assume (1). Let $x \in |\mathcal{X}|$ be a finite type point. We will produce a scheme over $\mathcal{X}$ which "works" in a neighbourhood of x. At the end of the proof we will take the disjoint union of all of these to conclude.

By Lemma lemma-point-finite-type-monomorphism the residual gerbe $\mathcal{Z}_x$ of $\mathcal{X}$ at x exists and $\mathcal{Z}_x \to \mathcal{X}$ is locally of finite type. By Lemma lemma-separation-properties-residual-gerbe the algebraic stack $\mathcal{Z}_x$ is DM. By Lemma lemma-DM-residual-gerbe there exists a field k and a surjective étale morphism $z : \mathrm{Spec}(k) \to \mathcal{Z}_x$. In particular the composition $x : \mathrm{Spec}(k) \to \mathcal{X}$ is locally of finite type (by Morphisms of Spaces, Lemmas spaces-morphisms-lemma-composition-finite-type and spaces-morphisms-lemma-etale-locally-finite-type).

Pick a scheme U and a smooth morphism $U \to \mathcal{X}$ such that x is in the image of $|U| \to |\mathcal{X}|$. Consider the following fibre square

$$\begin{array}{ccc} U & \longleftarrow & F \\ \downarrow & & \downarrow \\ \mathcal{X} & \xleftarrow{x} & \mathrm{Spec}(k) \end{array}$$

in other words $F = U \times_{\mathcal{X},x} \mathrm{Spec}(k)$. By Properties of Stacks, Lemma stacks-properties-lemma-points-cartesian we see that F is nonempty. As $\mathcal{Z}_x \to \mathcal{X}$ is a monomorphism we have

$$\mathrm{Spec}(k) \times_{z,\mathcal{Z}_x,z} \mathrm{Spec}(k) = \mathrm{Spec}(k) \times_{x,\mathcal{X},x} \mathrm{Spec}(k)$$

with étale projection maps to $\mathrm{Spec}(k)$ by construction of z. Since

$$F \times_U F = (\mathrm{Spec}(k) \times_{\mathcal{X}} \mathrm{Spec}(k)) \times_{\mathrm{Spec}(k)} F$$

we see that the projections maps $F \times_U F \to F$ are étale as well. It follows that $\Delta_{F/U} : F \to F \times_U F$ is étale (see Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-permanence). By Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open this implies that $\Delta_{F/U}$ is an open immersion, which finally implies by Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-unramified-morphism that $F \to U$ is unramified.

Pick a nonempty affine scheme V and an étale morphism $V \to F$. (This could be avoided by working directly with F, but it seems easier to explain what's going on by doing so.) Picture

$$\begin{array}{ccccc} U & \longleftarrow & F & \longleftarrow & V \\ \downarrow & & \downarrow & \swarrow & \\ \mathcal{X} & \xleftarrow{x} & \mathrm{Spec}(k) & & \end{array}$$

Then $V \to \mathrm{Spec}(k)$ is a smooth morphism of schemes and $V \to U$ is an unramified morphism of schemes (see Morphisms of Spaces, Lemmas spaces-morphisms-lemma-composition-smooth and spaces-morphisms-lemma-composition-unramified). Pick a closed point $v \in V$ with $k \subset \kappa(v)$ finite separable, see Varieties, Lemma varieties-lemma-smooth-separable-closed-points-dense. Let $u \in U$ be the image point. The local ring $\mathcal{O}_{V,v}$ is regular (see Varieties, Lemma varieties-lemma-smooth-regular) and the local ring homomorphism

$$\varphi : \mathcal{O}_{U,u} \longrightarrow \mathcal{O}_{V,v}$$

coming from the morphism $V \to U$ is such that $\varphi(\mathfrak{m}_u)\mathcal{O}_{V,v} = \mathfrak{m}_v$, see Morphisms, Lemma morphisms-lemma-unramified-at-point. Hence we can find $f_1, \ldots, f_d \in \mathcal{O}_{U,u}$ such that the images $\varphi(f_1), \ldots, \varphi(f_d)$ form a basis for $\mathfrak{m}_v/\mathfrak{m}_v^2$ over $\kappa(v)$. Since $\mathcal{O}_{V,v}$ is a regular local ring this implies that $\varphi(f_1), \ldots, \varphi(f_d)$ form a regular sequence in $\mathcal{O}_{V,v}$ (see Algebra, Lemma algebra-lemma-regular-ring-CM). After replacing U by an open neighbourhood of u we may assume $f_1, \ldots, f_d \in \Gamma(U, \mathcal{O}_U)$. After replacing U by a possibly even smaller open neighbourhood of u we may assume that $V(f_1, \ldots, f_d) \to \mathcal{X}$ is flat and locally of finite presentation, see Lemma lemma-slice. By construction

$$V(f_1, \ldots, f_d) \times_{\mathcal{X}} \mathrm{Spec}(k) \longleftarrow V(f_1, \ldots, f_d) \times_U V$$

is étale and $V(f_1, \ldots, f_d) \times_U V$ is the closed subscheme $T \subset V$ cut out by $f_1|_V, \ldots, f_d|_V$. Hence by construction $v \in T$ and

$$\mathcal{O}_{T,v} = \mathcal{O}_{V,v}/(\varphi(f_1), \ldots, \varphi(f_d)) = \kappa(v)$$

a finite separable extension of k. It follows that $T \to \mathrm{Spec}(k)$ is unramified at v, see Morphisms, Lemma morphisms-lemma-unramified-at-point. By definition of an unramified morphism of algebraic spaces this means that $V(f_1, \ldots, f_d) \times_{\mathcal{X}} \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is unramified at the image of v in $V(f_1, \ldots, f_d) \times_{\mathcal{X}} \mathrm{Spec}(k)$. Applying Lemma lemma-etale-at-point we see that on shrinking U to yet another open neighbourhood of u the morphism $V(f_1, \ldots, f_d) \to \mathcal{X}$ is étale.

We conclude that for every finite type point x of $\mathcal{X}$ there exists an étale morphism $f_x : W_x \to \mathcal{X}$ with x in the image of $|f_x|$. Set $W = \coprod_x W_x$ and $f = \coprod f_x$. Then f is étale. In particular the image of $|f|$ is open, see Properties of Stacks, Lemma stacks-properties-lemma-topology-points. By construction the image contains all finite type points of $\mathcal{X}$, hence f is surjective by Lemma lemma-enough-finite-type-points (and Properties of Stacks, Lemma stacks-properties-lemma-characterize-surjective). □